

KAGOSHIMA, Japan

WMO: 478270

Lat: 31.5550N

Lon: 130.5481E

Elev: 104

StdP: 14.64

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 34.8	(c) 36.8	(d) 19.9	(e) 15.0	(f) 44.3	(g) 22.5	(h) 17.0	(i) 43.6	(j) 20.6	(k) 48.6	(l) 18.9	(m) 48.4	(n) 7.9	(o) 340	(p) 0.398

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 10.6	(c) 92.4	(d) 78.8	(e) 90.9	(f) 78.4	(g) 89.4	(h) 78.1	(i) 80.9	(j) 88.3	(k) 80.1	(l) 87.2	(m) 79.4	(n) 86.3	(o) 10.1	(p) 280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.0	151.2	84.8	78.2	146.8	84.3	77.4	143.0	84.0	44.7	88.5	43.8	87.6	43.0	86.4	84.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 20.0	(b) 17.2	(c) 15.2	(e) 30.8	(f) 95.3	(g) 2.3	(h) 1.5	(i) 29.1	(j) 96.4	(k) 27.8	(l) 97.2	(m) 26.5	(n) 98.0	(o) 24.8	(p) 99.1		
			DB													
			WB													
				30.8	95.3	2.3	1.5	29.1	96.4	27.8	97.2	26.5	98.0	24.8	99.1	
				27.5	82.0	2.0	1.2	26.0	82.8	24.9	83.5	23.7	84.2	22.2	85.0	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	66.3	47.7	50.1	55.2	63.0	70.4	75.7	83.4	84.6	79.9	71.6	61.3	51.8
	DBStd	13.52	5.28	6.02	5.47	4.88	3.34	3.87	3.14	2.52	3.86	4.90	5.98	5.93
	HDD50	237	108	65	15	0	0	0	0	0	0	0	1	48
	HDD65	1912	536	416	306	93	3	0	0	0	0	7	141	410
	CDD50	6193	37	69	176	389	631	771	1035	1074	898	668	341	104
	CDD65	2393	0	0	2	32	169	321	570	609	448	210	31	1
	CDH74	22055	0	0	2	55	568	1966	6599	7492	4308	1021	44	0
	CDH80	7739	0	0	0	2	41	391	2614	3240	1322	128	1	0
(14) Wind	WSAvg	7.4	7.7	7.9	8.0	7.5	7.0	6.6	7.2	7.4	7.3	7.5	7.2	7.7
(15) Precipitation	PrecAvg	90.4	3.3	4.1	6.3	8.5	9.3	18.2	12.3	8.6	8.7	4.2	3.9	3.2
	PrecMax	158.5	7.9	9.8	13.8	17.8	22.0	33.8	41.5	24.8	20.9	14.2	8.5	9.6
	PrecMin	60.0	0.4	1.2	1.3	1.6	2.2	7.8	0.7	0.9	1.1	0.4	1.0	0.3
	PrecStd	17.9	1.7	2.2	2.9	4.1	3.9	7.2	7.0	5.1	4.9	3.2	2.0	2.0
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	65.9	69.4	72.5	78.5	83.4	89.4	94.2	94.9	91.7	85.9	78.0	69.9
		MCWB	59.5	62.7	61.6	65.6	67.5	78.1	79.0	79.4	77.4	74.2	68.4	61.6
	2%	DB	62.0	66.0	69.3	75.3	81.0	86.4	92.3	92.9	89.5	83.3	75.0	66.6
		MCWB	55.7	60.0	59.8	63.8	67.4	76.8	78.6	79.2	76.7	72.1	65.9	59.9
	5%	DB	59.4	63.0	66.9	73.3	79.1	83.9	90.8	91.4	87.8	81.1	72.9	64.2
		MCWB	52.9	56.7	58.5	62.6	67.1	75.2	78.5	78.7	76.0	70.7	64.7	57.1
	10%	DB	57.1	60.3	64.6	71.2	77.2	82.0	89.2	89.9	86.2	79.2	70.9	61.7
		MCWB	50.8	53.8	57.1	61.9	66.3	74.0	78.3	78.3	75.4	69.4	63.2	55.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	61.7	64.6	65.7	70.9	74.7	79.3	81.9	82.5	80.4	77.3	71.7	65.0
		MCDB	64.7	67.5	69.0	74.9	79.2	85.2	89.0	90.2	86.8	82.4	76.0	67.1
	2%	WB	58.0	61.4	62.9	68.2	72.7	78.3	80.8	81.2	79.2	75.5	68.9	61.6
		MCDB	60.8	65.0	66.4	71.8	76.5	83.9	88.1	88.7	85.5	80.8	72.6	65.2
	5%	WB	54.6	58.3	60.9	66.1	71.0	77.3	79.9	80.4	78.3	73.7	66.8	58.8
		MCDB	58.1	62.1	65.2	70.1	74.9	82.6	87.1	87.8	84.6	79.3	71.0	63.2
	10%	WB	51.3	55.1	58.6	64.3	69.3	76.2	79.2	79.7	77.5	71.5	64.6	56.2
		MCDB	56.1	59.1	63.5	68.8	74.1	81.0	86.1	86.9	83.7	77.1	69.6	60.9
(35) Mean Daily Temperature Range	5% DB	MDBR	13.3	14.1	14.6	14.4	13.3	9.8	10.5	10.6	11.2	12.6	13.4	13.6
		MCDBR	16.1	15.4	16.7	16.7	16.1	11.9	12.6	12.4	12.3	13.2	13.5	15.1
	5% WB	MCWBR	11.5	11.3	10.6	9.0	7.8	5.1	4.3	4.3	5.0	6.5	7.9	10.3
		MCWBR	13.7	13.2	13.5	12.0	10.5	9.7	11.2	10.9	10.2	10.0	10.0	12.5
(40) Clear-Sky Solar Irradiance	taub		0.428	0.465	0.521	0.539	0.546	0.546	0.527	0.517	0.502	0.469	0.431	0.419
	taud		2.102	2.031	1.892	1.902	1.951	2.040	2.127	2.155	2.150	2.159	2.188	2.135
	Ebn at Noon		244	247	242	244	243	242	246	247	246	244	242	239
	Edh at Noon		42	49	60	62	60	54	49	48	46	43	38	38
(44) All-Sky Solar Radiation	RadAvg		757	963	1291	1515	1574	1284	1696	1713	1427	1190	881	718
	RadStd		70	85	89	151	177	204	191	120	148	133	84	71

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	+0.72	N/A	N/A	N/A	N/A
(47) Regional (20 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page